

d) "Every student has an Internet account." "Homer does not have an Internet account." "Maggie has an Internet account."

[Soln:] $S(x)$: x is a Student $I(x)$: x has an Internet account

[Premise:] 1) $\forall x(S(x) \Rightarrow I(x))$ 2) $\neg I(\text{Homer})$ 3) $I(\text{Maggie})$

Steps

1. $\forall x(S(x) \Rightarrow I(x))$

Reason

Premise

2. $\neg I(\text{Homer})$

Premise

3. $I(\text{Maggie})$

Premise

4. $S(\text{Homer}) \Rightarrow I(\text{Homer})$

Universal Instantiation on 1

5. $\neg S(\text{Homer})$

Modus tollens on 2, 4

[Valid conclusion:] Homer is not a Student

e) "All foods that are healthy to eat do not taste good." "Tofu is healthy to eat." "You only eat what tastes good." "You do not eat tofu." "Cheeseburgers are not healthy to eat." $E(x)$: You eat food x $T(x)$: Food x tastes good.

[Soln:] $H(x)$: Food x is Healthy to eat $T(x)$: Food x tastes good

[Premise:] 1) $\forall x(H(x) \Rightarrow \neg T(x))$ 2) $H(\text{Tofu})$ 3) $\forall x(E(x) \Rightarrow T(x))$

4) $\neg E(\text{Tofu})$ 5) $\neg H(\text{cheeseburgers})$

Reason

Steps

1. $\forall x(H(x) \Rightarrow \neg T(x))$

Premise

2. $H(\text{Tofu})$

Premise

3. $\forall x(E(x) \Rightarrow T(x))$

Premise

4. $\neg E(\text{Tofu})$

Premise

5. $\neg H(\text{cheeseburgers})$

Premise

6. $H(\text{Tofu}) \Rightarrow \neg T(\text{Tofu})$

Universal instantiation on 1

7. $\neg T(\text{Tofu})$

Modus ponens on 2, 6